

SWAGGER UI

Board Game Café Management System1.0.0GA1.1

/api/swagger.json

A MongoDB + FastAPI project to manage a board game café: categories, games, tables, and play sessions.

Categories		^
GET	/categories/ Get All Categories	▼
POST	/categories/ Create Category	▼
GET	/categories/{category_id} Get Category	▼
PUT	/categories/{category_id} Update Category	▼
DELETE	/categories/{category_id} Delete Category	▼
Games		^
GET	/games/ Get All Games	▼
POST	/games/ Create Game	▼
GET	/games/{game_id} Get Game	▼
PUT	/games/{game_id} Update Game	▼
DELETE	/games/{game_id} Delete Game	▼
Tables		^
GET	/tables/ Get All Tables	▼
POST	/tables/ Create Table	▼
GET	/tables/{table_id} Get Table	▼
PUT	/tables/{table_id} Update Table	▼
DELETE	/tables/{table_id} Delete Table	▼
Sessions		^
GET	/sessions/ Get All Sessions	▼
GET	/sessions/{session_id} Get Session	▼
DELETE	/sessions/{session_id} Delete Session	▼
POST	/sessions/start Start Session	▼
POST	/sessions/end/{session_id} End Session	▼
POST	/sessions/{session_id}/add-game Add Game To Session	▼
GET	/sessions/details/{session_id} Get Session Details	▼
Root		^
GET	/ Root	▼
Reports		^
GET	/reports/dashboard Dashboard Report	▼
GET	/reports/session-history Session History	▼
GET	/reports/most-played-games Most Played Games	▼
GET	/reports/most-used-tables Most Used Tables	▼

SCHEMAS

Schemas			^
AddGameRequest	>	Expand all	object
CategoryCreate	>	Expand all	object
CategoryResponse	>	Expand all	object
CategoryUpdate	>	Expand all	object
GameCreate	>	Expand all	object
GameResponse	>	Expand all	object
GameUpdate	>	Expand all	object
HTTPValidationError	>	Expand all	object
SessionResponse	>	Expand all	object
SessionStartRequest	>	Expand all	object
TableCreate	>	Expand all	object
TableResponse	>	Expand all	object
TableUpdate	>	Expand all	object
ValidationError	>	Expand all	object

Category Creation

Categories

GET /categories/ Get All Categories

POST /categories/ Create Category

Create a new category.

Steps:

1. Check that category_id is not already used.

2. Insert the new category document into MongoDB.

Parameters

No parameters

CancelReset

Request body required

application/json

Edit Value | Schema

```
{  "category_id": "CAT001",  "name": "Strategy",  "description": "Strategic planning and decision making games"}

```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/categories/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "category_id": "CAT001",
    "name": "Strategy",
    "description": "Strategic planning and decision making games"
  }'
```

Request URL

```
http://127.0.0.1:8080/categories/
```

Server response

Code	Details
200	Response body <pre>{ "category_id": "CAT001", "name": "Strategy", "description": "Strategic planning and decision making games"} </pre> Download Response headers <pre>content-length: 103 content-type: application/json date: Thu, 18 Jun 2026 13:37:35 GMT server: uvicorn </pre>

Responses

Code	Description	Links
200	Successful Response	No links

Game Creation

Games

GET /games/ Get All Games

POST /games/ Create Game

Create a new game.
Steps:
1. Check that game_id is not already used.
2. Check that the referenced category_id actually exists (this enforces the Games -> Category relationship).
3. Insert the new game document.

Parameters

CancelReset

No parameters

Request body required

application/json

Edit Value | Schema

```
{  "game_id": "GAME010",  "name": "Cards Against Humanity",  "category_id": "CAT002",  "min_players": 4,  "max_players": 12,  "status": "Available"}
```

ExecuteClear

Responses

Curl

```
curl -X 'POST' \  'http://127.0.0.1:8000/games/' \  -H 'accept: application/json' \  -H 'Content-Type: application/json' \  -d '{  "game_id": "GAME010",  "name": "Cards Against Humanity",  "category_id": "CAT002",  "min_players": 4,  "max_players": 12,  "status": "Available"  }'
```

Request URL

http://127.0.0.1:8000/games/

Server response

CodeDetails

200

Response body

```
{  "game_id": "GAME010",  "name": "Cards Against Humanity"
```

Session Start

Sessions

GET

/sessions/ Get All Sessions

⌵

GET

/sessions/{session_id} Get Session

⌵

DELETE

/sessions/{session_id} Delete Session

⌵

POST

/sessions/start Start Session

⌵

Start a new session.

- Validates table and games.
- Creates the session.
- Marks table as Occupied.
- Marks games as Being Played.

See services/session_service.py for the full logic.

Parameters

Cancel

Reset

No parameters

Request body required

application/json ⌵

Edit Value | Schema

```
{  "session_id": "SESSION002",  "group_name": "College Legends",  "table_id": "TABLE001",  "game_ids": [    "GAME003",    "GAME006"  ],  "player_count": 5}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  "http://127.0.0.1:8000/sessions/start" \  -H 'accept: application/json' \  -H 'Content-Type: application/json' \  -d '{  "session_id": "SESSION002",  "group_name": "College Legends",  "table_id": "TABLE001",  "game_ids": [    "GAME003",    "GAME006"  ],  "player_count": 5}
```

Request URL

http://127.0.0.1:8000/sessions/start

Server response

Code

Details

200

Response body

```
{  "session_id": "SESSION002",  "group_name": "College Legends",  "table_id": "TABLE001",  "game_ids": [    "GAME003",    "GAME006"  ],  "player_count": 5}
```

Table Creation

Tables

GET /tables/ Get All Tables

POST /tables/ Create Table

Create a new table. table_id must be unique.

Parameters

No parameters

Request body required

application/json

Cancel Reset

Edit Value Schema

```
{  "table_id": "TABLE005",  "name": "Castle Table",  "capacity": 10,  "status": "Available"}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/tables/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "table_id": "TABLE005",
    "name": "Castle Table",
    "capacity": 10,
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/tables/

GET

- CATEGORIES

```
127.0.0.1:8000/categories/

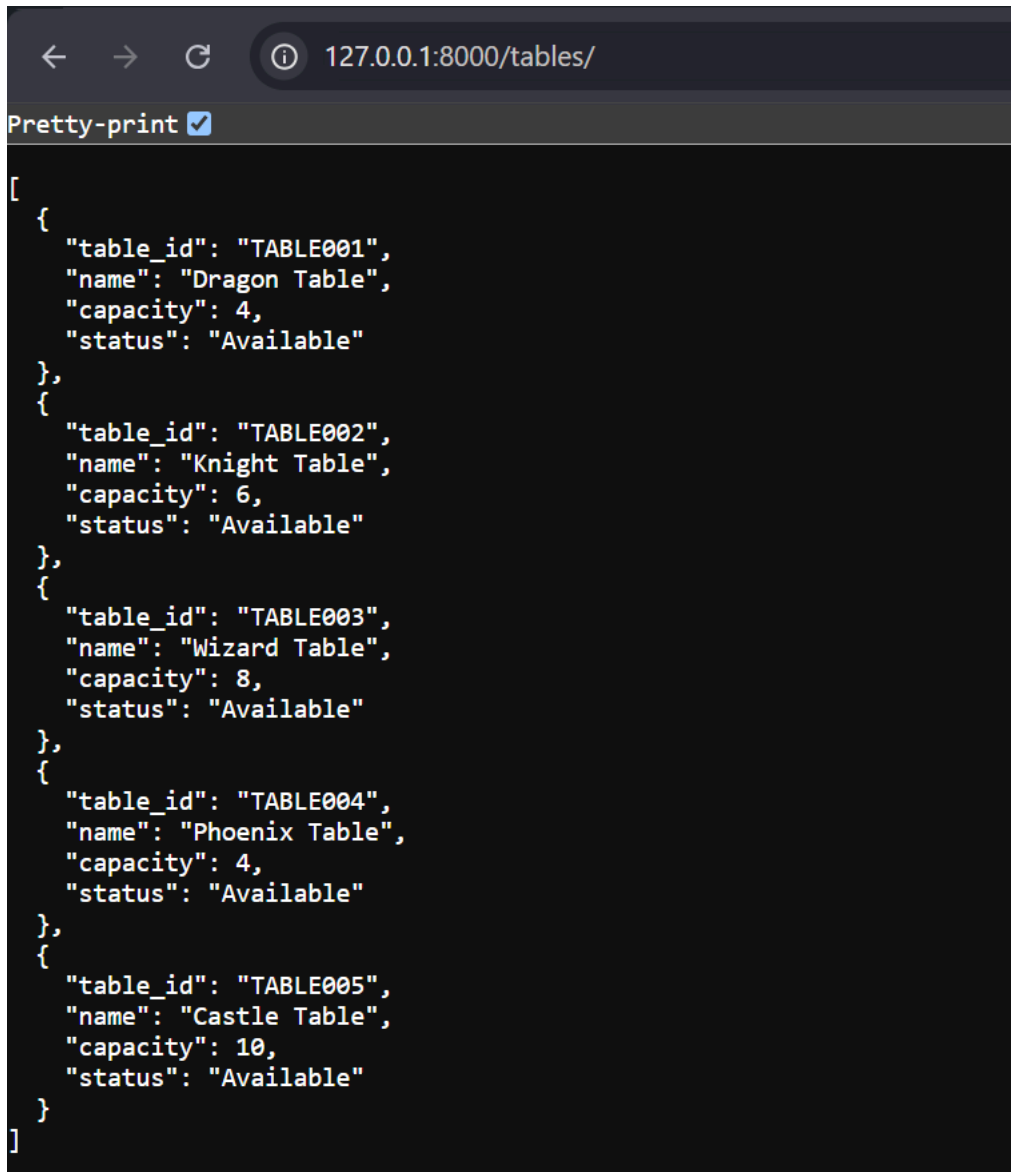
[
  {
    "category_id": "CAT002",
    "name": "Party",
    "description": "Party games"
  },
  {
    "category_id": "CAT003",
    "name": "Card",
    "description": "Card based games"
  },
  {
    "category_id": "CAT004",
    "name": "Family",
    "description": "Family-friendly board games"
  },
  {
    "category_id": "CAT005",
    "name": "Puzzle",
    "description": "Logic and puzzle solving games"
  },
  {
    "category_id": "CAT001",
    "name": "Strategy",
    "description": "Strategic planning and decision making games"
  }
]
```

- GAMES

```
← → ↺ ⓘ 127.0.0.1:8000/games/

{
  {
    "game_id": "GAME001",
    "name": "Catan",
    "category_id": "CAT001",
    "min_players": 3,
    "max_players": 4,
    "status": "Available"
  },
  {
    "game_id": "GAME002",
    "name": "Chess",
    "category_id": "CAT001",
    "min_players": 2,
    "max_players": 2,
    "status": "Available"
  },
  {
    "game_id": "GAME003",
    "name": "Risk",
    "category_id": "CAT001",
    "min_players": 2,
    "max_players": 6,
    "status": "Available"
  },
  {
    "game_id": "GAME004",
    "name": "Monopoly",
    "category_id": "CAT004",
    "min_players": 2,
    "max_players": 6,
    "status": "Available"
  },
  {
    "game_id": "GAME005",
    "name": "Uno",
    "category_id": "CAT003",
    "min_players": 2,
    "max_players": 10,
    "status": "Available"
  },
  {
    "game_id": "GAME006",
    "name": "Ludo",
    "category_id": "CAT004",
    "min_players": 2,
    "max_players": 6,
    "status": "Being Played"
  },
  {
    "game_id": "GAME007",
    "name": "Scrabble",
    "category_id": "CAT004",
    "min_players": 2,
    "max_players": 4,
    "status": "Available"
  },
  {
    "game_id": "GAME008",
    "name": "Sudoku Challenge",
    "category_id": "CAT005",
    "min_players": 1,
    "max_players": 4,
    "status": "Available"
  },
  {
    "game_id": "GAME009",
    "name": "Ticket to Ride",
    "category_id": "CAT001",
    "min_players": 2,
    "max_players": 5,
    "status": "Available"
  },
  {
    "game_id": "GAME010",
    "name": "Cards Against Humanity",
    "category_id": "CAT002",
    "min_players": 4,
    "max_players": 12,
    "status": "Available"
  }
}
```

- TABLES



A screenshot of a web browser window. The address bar shows the URL `127.0.0.1:8000/tables/`. Below the address bar, there is a toggle for "Pretty-print" which is checked. The main content area displays a JSON array of five table objects. Each object contains the following fields: `table_id`, `name`, `capacity`, and `status`. The tables are: "Dragon Table" (capacity 4), "Knight Table" (capacity 6), "Wizard Table" (capacity 8), "Phoenix Table" (capacity 4), and "Castle Table" (capacity 10). All tables have a status of "Available".

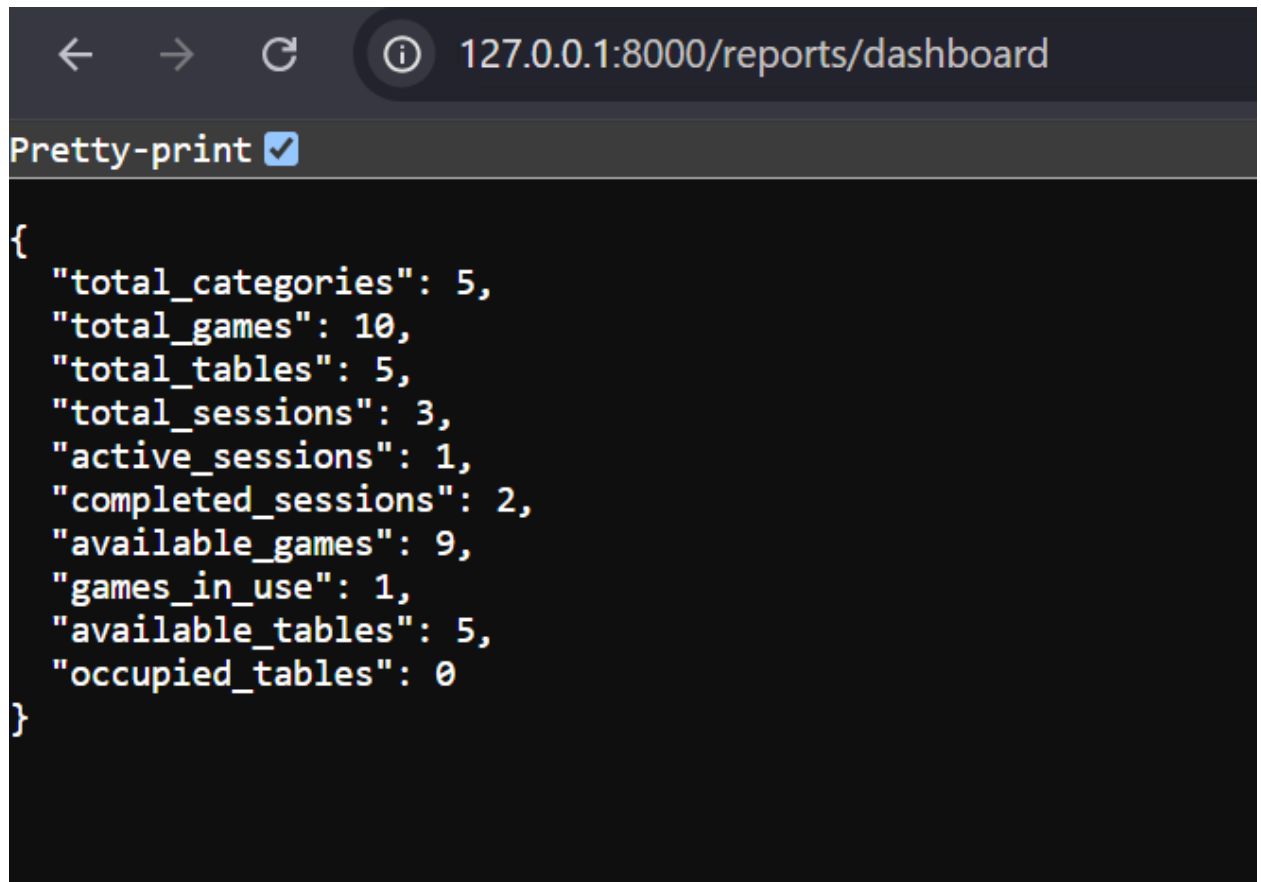
```
[
  {
    "table_id": "TABLE001",
    "name": "Dragon Table",
    "capacity": 4,
    "status": "Available"
  },
  {
    "table_id": "TABLE002",
    "name": "Knight Table",
    "capacity": 6,
    "status": "Available"
  },
  {
    "table_id": "TABLE003",
    "name": "Wizard Table",
    "capacity": 8,
    "status": "Available"
  },
  {
    "table_id": "TABLE004",
    "name": "Phoenix Table",
    "capacity": 4,
    "status": "Available"
  },
  {
    "table_id": "TABLE005",
    "name": "Castle Table",
    "capacity": 10,
    "status": "Available"
  }
]
```

- SESSIONS



```
[
  {
    "session_id": "SESSION001",
    "group_name": "Weekend Gamers",
    "table_id": "TABLE001",
    "game_ids": [
      "GAME001",
      "GAME002"
    ],
    "player_count": 4,
    "status": "Completed",
    "start_time": "2026-06-17T18:06:49.967000",
    "end_time": "2026-06-17T18:13:21.712000"
  },
  {
    "session_id": "SESSION002",
    "group_name": "College Legends",
    "table_id": "TABLE002",
    "game_ids": [
      "GAME003",
      "GAME006"
    ],
    "player_count": 5,
    "status": "Completed",
    "start_time": "2026-06-18T13:42:30.441000",
    "end_time": "2026-06-18T13:49:43.210000"
  },
  {
    "session_id": "SESSION005",
    "group_name": "Family Fun Group",
    "table_id": "TABLE005",
    "game_ids": [
      "GAME006"
    ],
    "player_count": 4,
    "status": "Active",
    "start_time": "2026-06-18T13:50:07.847000",
    "end_time": null
  }
]
```

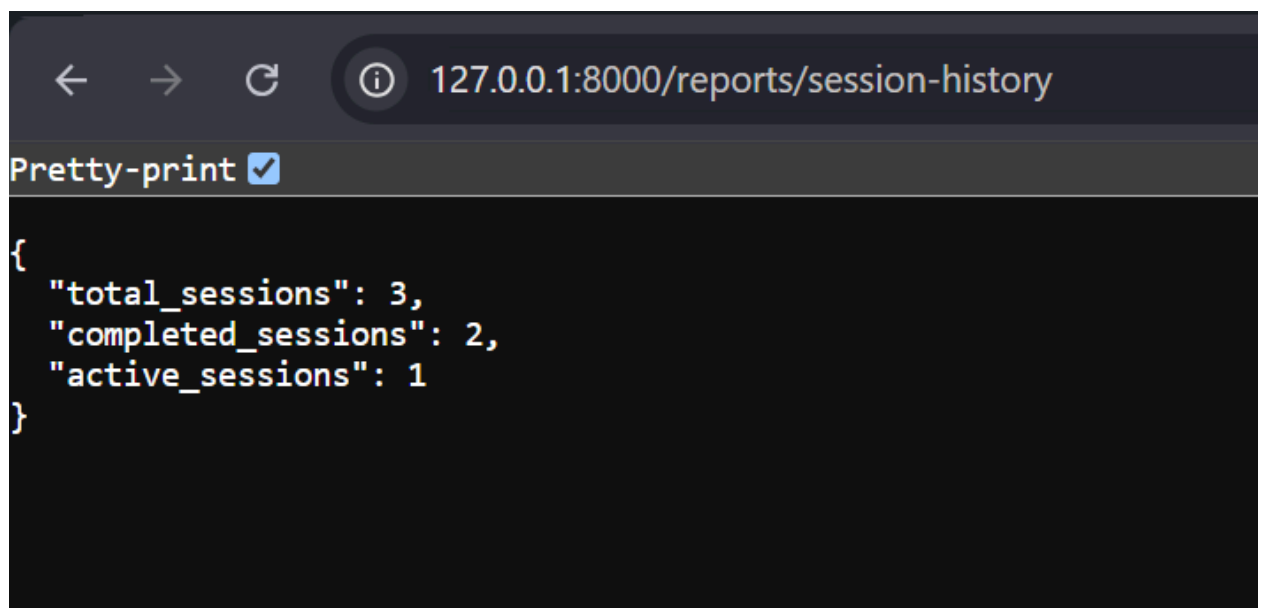
DASHBOARD



A screenshot of a web browser displaying a dashboard report. The address bar shows the URL `127.0.0.1:8000/reports/dashboard`. Below the address bar, there is a toggle for "Pretty-print" which is checked. The main content area displays a JSON object with the following data:

```
{
  "total_categories": 5,
  "total_games": 10,
  "total_tables": 5,
  "total_sessions": 3,
  "active_sessions": 1,
  "completed_sessions": 2,
  "available_games": 9,
  "games_in_use": 1,
  "available_tables": 5,
  "occupied_tables": 0
}
```

SESSION HISTORY



A screenshot of a web browser displaying a session history report. The address bar shows the URL `127.0.0.1:8000/reports/session-history`. Below the address bar, there is a toggle for "Pretty-print" which is checked. The main content area displays a JSON object with the following data:

```
{
  "total_sessions": 3,
  "completed_sessions": 2,
  "active_sessions": 1
}
```

MOST PLAYED GAMES



The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8000/reports/most-played-games`. Below the address bar, there is a toggle for 'Pretty-print' which is checked. The main content area displays a JSON array of game objects, formatted for readability. The JSON data is as follows:

```
[
  {
    "game_id": "GAME006",
    "game_name": "Ludo",
    "times_played": 2
  },
  {
    "game_id": "GAME003",
    "game_name": "Risk",
    "times_played": 1
  },
  {
    "game_id": "GAME001",
    "game_name": "Catan",
    "times_played": 1
  },
  {
    "game_id": "GAME002",
    "game_name": "Chess",
    "times_played": 1
  }
]
```

MOST USED TABLES

```
127.0.0.1:8000/reports/most-used-tables

Pretty-print ☒

[
  {
    "table_id": "TABLE005",
    "table_name": "Castle Table",
    "sessions_hosted": 1
  },
  {
    "table_id": "TABLE001",
    "table_name": "Dragon Table",
    "sessions_hosted": 1
  },
  {
    "table_id": "TABLE002",
    "table_name": "Knight Table",
    "sessions_hosted": 1
  }
]
```

MONGODB COMPASS UI

Welcome > mongosh: localhost:27017 boardgame_cafe

games tables categories sessions +

localhost:27017 > boardgame_cafe

Open MongoDB shell Create collection Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
categories	-	32.77 kB	532.00 B	5	106.00 B	1	32.77 kB
games	-	36.86 kB	1.43 kB	10	143.00 B	1	36.86 kB
sessions	-	32.77 kB	442.00 B	2	221.00 B	1	36.86 kB
tables	-	32.77 kB	417.00 B	4	104.00 B	1	36.86 kB

GAMES DATABASE

Welcome🔍 mongoosh: localhost:27017boardgame_cafe

games

tables

categories

sessions

+

localhost:27017 > boardgame_cafe > games

Open MongoDB shell

Documents10AggregationsSchemaIndexes1Validation

🔍 {}

Generate query⚡️

Explain

Reset

Find

Options➤

➕ ADD DATA

📄 BULK

📄 EXPORT DATA

📄 EXPORT CODE

100

1 - 10 of 10

↺

↻

↷

☰

🔍

📄

_id: ObjectId('6a30344bf1df7f0c3c283aef')

game_id: "GAME001"

name: "Catan"

category_id: "CAT001"

min_players: 3

max_players: 4

status: "Available"

_id: ObjectId('6a303459f1df7f0c3c283afe')

game_id: "GAME002"

name: "Chess"

category_id: "CAT001"

min_players: 2

max_players: 2

status: "Available"

_id: ObjectId('6a32eb34f609c5cd0aa2f1ce')

game_id: "GAME003"

name: "Risk"

category_id: "CAT001"

min_players: 2

max_players: 6

status: "Available"

_id: ObjectId('6a32eb4bf609c5cd0aa2f1cf')

game_id: "GAME004"

name: "Monopoly"

category_id: "CAT004"

min_players: 2

max_players: 6

status: "Available"

_id: ObjectId('6a32eb5bf609c5cd0aa2f1d0')

game_id: "GAME005"

name: "Uno"

category_id: "CAT003"

min_players: 2

max_players: 10

status: "Available"

Welcome🔍 mongoosh: localhost:27017boardgame_cafe

games

tables

categories

sessions

+

localhost:27017 > boardgame_cafe > games

Open MongoDB shell

Documents10AggregationsSchemaIndexes1Validation

🔍 {}

Generate query⚡️

Explain

Reset

Find

Options➤

➕ ADD DATA

📄 BULK

📄 EXPORT DATA

📄 EXPORT CODE

100

1 - 10 of 10

↺

↻

↷

☰

🔍

📄

max_players: 10

status: "Available"

_id: ObjectId('6a32eb76f609c5cd0aa2f1d1')

game_id: "GAME006"

name: "Ludo"

category_id: "CAT004"

min_players: 2

max_players: 6

status: "Being Played"

_id: ObjectId('6a32eb84f609c5cd0aa2f1d2')

game_id: "GAME007"

name: "Scrabble"

category_id: "CAT004"

min_players: 2

max_players: 4

status: "Available"

_id: ObjectId('6a32eb8ef609c5cd0aa2f1d3')

game_id: "GAME008"

name: "Sudoku Challenge"

category_id: "CAT005"

min_players: 1

max_players: 4

status: "Available"

_id: ObjectId('6a32eb98f609c5cd0aa2f1d4')

game_id: "GAME009"

name: "Ticket to Ride"

category_id: "CAT001"

min_players: 2

max_players: 5

status: "Available"

_id: ObjectId('6a33f558aa82a28c180138ac')

game_id: "GAME010"

name: "Cards Against Humanity"

category_id: "CAT002"

min_players: 4

max_players: 12

status: "Available"

TABLES DATABASE-

localhost:27017 > boardgame_cafe > tables

Documents 4 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 4 of 4

```
{
  "_id": {},
  "table_id": "TABLE001",
  "name": "Dragon Table",
  "capacity": 4,
  "status": "Available"
}
```

```
{
  "_id": {},
  "table_id": "TABLE002",
  "name": "Knight Table",
  "capacity": 6,
  "status": "Available"
}
```

```
{
  "_id": {},
  "table_id": "TABLE003",
  "name": "Wizard Table",
  "capacity": 8,
  "status": "Available"
}
```

```
{
  "_id": {},
  "table_id": "TABLE004",
  "name": "Phoenix Table",
  "capacity": 4,
  "status": "Available"
}
```

SESSIONS DATABASE-

mongodb:localhost:27017 > boardgame_cafe > sessions

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT DATA](#) [EXPORT CODE](#) 25 1 - 3 of 3

```

{
  "_id": ObjectId("6a32e2391df7f0c3c283af3"),
  "session_id": "SESSION001",
  "group_name": "Weekend Gamers",
  "table_id": "TABLE001",
  "game_ids": Array (2)
  "player_count": 4
  "status": "Completed"
  "start_time": 2026-06-17T18:06:49.967+00:00
  "end_time": 2026-06-17T18:13:21.712+00:00
}

```

```

{
  "_id": ObjectId("6a33f5c6aa82a28c180138ad"),
  "session_id": "SESSION002",
  "group_name": "College Legends",
  "table_id": "TABLE002",
  "game_ids": Array (2)
  "player_count": 5
  "status": "Completed"
  "start_time": 2026-06-18T13:42:30.441+00:00
  "end_time": 2026-06-18T13:49:43.210+00:00
}

```

```

{
  "_id": ObjectId("6a33f78faa82a28c180138ae"),
  "session_id": "SESSION005",
  "group_name": "Family Fun Group",
  "table_id": "TABLE005",
  "game_ids": Array (1)
  "player_count": 4
  "status": "Active"
  "start_time": 2026-06-18T13:50:07.847+00:00
  "end_time": null
}

```

CATEGORIES DATABASE-

Welcome

mongoosh:localhost:27017

boardgame_cafe

games

tables

categories

sessions

+

localhost:27017 > boardgame_cafe > categories

Documents 5

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#) +

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

251 - 5 of 5

▾

▸

"_id": {}

"category_id": "CAT002"

"name": "Party"

"description": "Party games"

▾

▸

"_id": {}

"category_id": "CAT003"

"name": "Card"

"description": "Card based games"

▾

▸

"_id": {}

"category_id": "CAT004"

"name": "Family"

"description": "Family-friendly board games"

▾

▸

"_id": {}

"category_id": "CAT005"

"name": "Puzzle"

"description": "Logic and puzzle solving games"

▾

▸

"_id": {}

"category_id": "CAT001"

"name": "Strategy"

"description": "Strategic planning and decision making games"